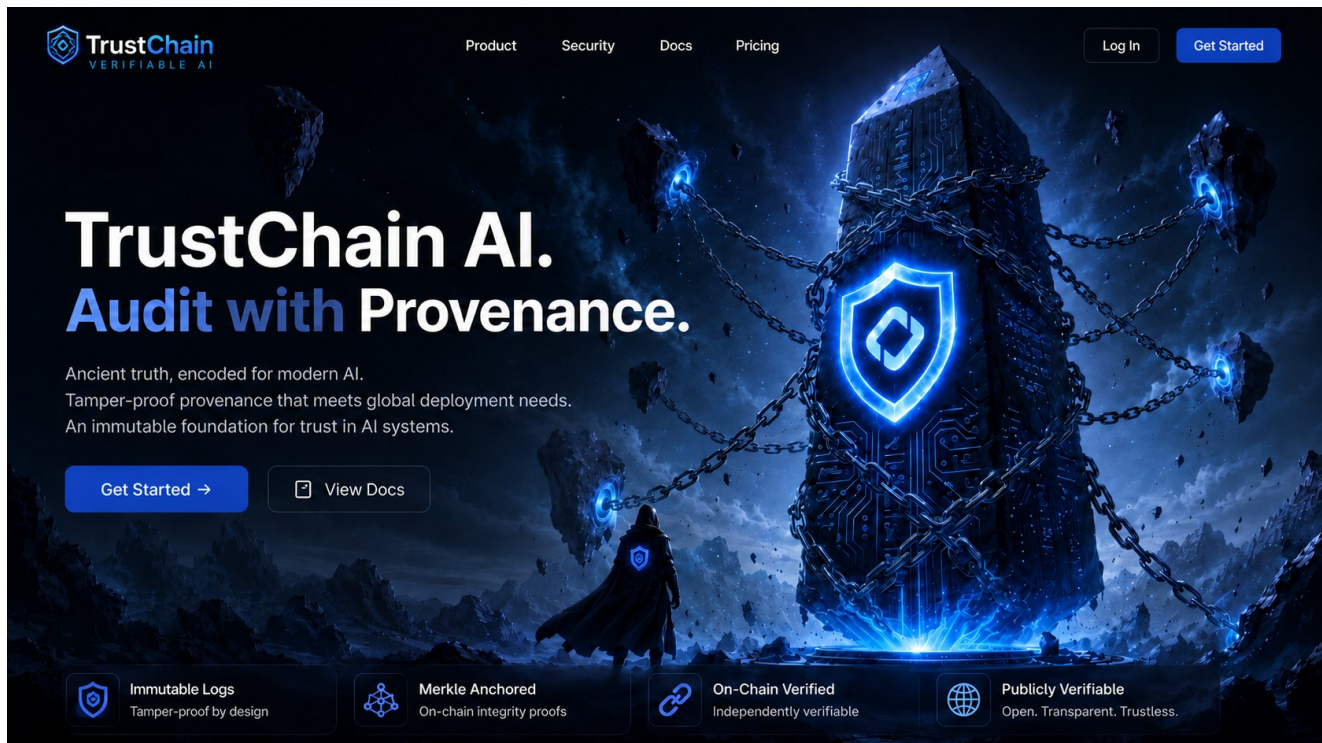


Cinematic Landing Page Master Implementation Specification

Version 3 - FINAL build prompt: punch-triggered awakening, dual experience modes, approved section copy

This version is intended to be handed directly to the coding agent. Approved Product, Security and four proof-feature section copy is integrated below.



Approved hero composition reference. Rebuild as a real interactive 3D scene; do not ship this as a flattened full-screen image.

1. Final product vision

TrustChain should open like an interactive sci-fi film, but remain a serious production website. The user does not merely watch an intro: the user's scroll causes the world to awaken. The first hero sequence is the most dramatic part of the site. After that, the landing page continues as a connected, cinematic product story.

Core interaction rule

SCROLL triggers major transformations. IDLE keeps the world subtly alive. CURSOR causes local physical or visual reactions. ENTER or ESC switches the visitor from the full cinematic experience into a faster normal website experience.

- Delete the existing landing-page UI implementation and rebuild the frontend presentation from scratch.
- Preserve backend, API, auth, routing, utilities and build infrastructure unless a change is genuinely required.
- Do not make two duplicated landing pages. Build one component/content structure with two experience modes.
- All important UI text remains real HTML/CSS above or beside WebGL. Do not render critical text into textures.
- The experience must feel smooth, intentional and expensive rather than chaotic or game-like.
- Approved copy rule: use the supplied section copy in this specification as the content source for Product, Security, Immutable Logs, Merkle Anchored, On-Chain Verified and Publicly Verifiable. Preserve the exact four pipeline labels Researcher / Validator / Scorer / Reporter because they match the running LangGraph node names. Do not rename those nodes.

2. TrustChain context

TrustChain is an AI auditability platform for multi-agent workflows. It records agent steps and cryptographically anchors audit data in Merkle batches on-chain so independent parties can verify integrity without trusting the application database. The visual metaphor is therefore containment, permanence, proof, energy and cryptographic anchoring.

Copy policy for this version

Do NOT invent or lock final body copy for Product, Immutable Logs, Merkle Anchoring, On-Chain Verification, Public Verification or Security. Build these sections so all copy comes from a centralized content configuration. Final section text will be provided separately in the next revision. Navigation labels, CTA labels and feature names may be used because they are interaction labels rather than final marketing copy.

3. Approved visual identity



Brand treatment

Use the cyan/blue shield-chain emblem, TrustChain wordmark and small uppercase VERIFIABLE AI subtitle in the navbar. The navbar brand is not visible as a normal static brand at the very beginning of the cinematic; it is born from the monolith and moves into the navbar during the opening sequence.

4. Two experience modes - one website

FULL CINEMATIC MODE (default)	QUICK / NORMAL MODE (ENTER or ESC)
User scroll drives a large directed 3D story.	User immediately reaches the awakened hero and browses a normal vertical website.
Pinned scroll sequences and major camera travel are allowed.	No major camera flights, giant scene travel or long pinned cinematic journeys.
Product sections transition into one another as a continuous 3D world.	The same sections become contained, premium 3D/DOM visualizations with entrance animations.
Movie-like blockchain world, camera journeys, morphing scene transitions.	Floating blockchain visualization remains, but camera stays restrained and reading stays easy.
All hero physics and ambient interaction remain active.	All hero physics and ambient interaction remain active.

- Default state: experienceMode = cinematic.
- During the opening only, pressing ENTER or ESC sets experienceMode = normal and fast-forwards the opening to the final hero state.
- After the opening has resolved, remove the global ENTER/ESC cinematic listeners so normal keyboard interaction works as expected.
- Do not store the mode forever. Session-level behavior is acceptable; a fresh session may offer the full cinematic again.

```
type ExperienceMode = "cinematic" | "normal";
```

```
default: "cinematic"
```

```
ENTER or ESC during opening:
```

```
-> resolve opening in ~300-500ms
```

```
-> set introAwakened = true
```

```
-> set experienceMode = "normal"
```

```
-> move internal scroll state to end of intro
```

```
-> keep hero physics + ambience alive
```

```
-> next scroll begins normal Product section
```

5. Opening affordance - no visible skip button

Bottom-center, extremely subtle:

SCROLL TO AWAKEN

v

ENTER / ESC FOR QUICK VIEW

- Do not add a visible Skip button.
- The hint must be small, low-opacity and secondary to the scene.
- SCROLL TO AWAKEN fades after the visitor begins scrolling.
- ENTER / ESC FOR QUICK VIEW remains faintly visible until the opening resolves or the user activates Quick View.
- If prefers-reduced-motion is enabled, start close to the awakened hero/normal-mode state and avoid intense shake.

6. Full opening cinematic - scroll controlled, not autoplay

The hero should occupy a long pinned scroll range. Scroll position controls the cinematic timeline with scrubbed progress. The percentages below are a conceptual guide, not fixed numbers. Tune them by feel after testing.

0% dark environment / camera follows the man from behind
8-18% man walks forward; wind/dust build; camera remains low and behind him
18-26% man slows, plants himself and prepares the punch
26-32% camera swings into a low three-quarter/side angle; man punches the ground
32-42% physical ground shockwave, cracks, debris and camera impulse propagate from the punch
42-60% monolith rises slowly from below the fractured ground; camera returns behind/beside the man and tracks upward
60-66% monolith settles; brief dark/silent anticipation
66-70% TrustChain core/logo ignites almost instantly
70-80% huge spherical blue energy wave expands through the world
74-84% rocks energize and levitate only as the spherical wave reaches them
82-92% only the approved reference chains deploy sequentially with full physics and bind the monolith
92-97% logo copy emerges from monolith and travels into the navbar
97-100% hero text, CTAs, four feature controls and compact nav resolve into the final living hero state

6.1 Character arrival

- The man/guardian appears first. The opening camera follows him from behind at a low cinematic height. The monolith must NOT already be standing in the final hero composition.
- Scroll advances the walk animation and camera follow. If scroll pauses, directed movement pauses while ambient wind, cloth and dust may continue.
- The man walks into position, slows, plants himself, winds up and punches the ground. After impact he braces against the eruption and finally looks upward as the monolith towers over him.
- Use framing and scale so the man feels small and the monolith feels colossal. The punch must feel like the causal trigger for the physical ground awakening.
- The cloak remains animated throughout the entire hero experience, including after the cinematic is finished or skipped.

6.1.1 Hero camera choreography - five meaningful shots

Do not constantly swing the camera. Use a few deliberate shots connected by smooth spline/orbit transitions so it feels like cinema rather than a game camera.

- SHOT 1 - Behind the man: low follow camera as he walks into the empty landscape; strong foreground parallax and wind-driven cloth.
- SHOT 2 - Punch setup/impact: camera smoothly orbits roughly 40-60 degrees into a low three-quarter/side view so the user clearly sees the man punch the ground.
- SHOT 3 - Consequence: camera returns behind/slightly beside the man as cracks race outward and the monolith begins rising in front of him.
- SHOT 4 - Monumental wide: frame the man small and the fully emerged monolith enormous for the instant core ignition and spherical energy-wave event.
- SHOT 5 - Final hero composition: settle into the approved landing-page composition with chains, floating rocks, living atmosphere and the UI reveal.

Scroll progress controls transitions between these shots. Scroll velocity may influence secondary effects, but camera interpolation must remain damped, deterministic and comfortable.

6.2 Punch impact, ground shockwave, wind and dust

- Wind begins subtle and builds as the character approaches the punch point. The opening should feel calm enough that the punch has contrast.
- Dust should move directionally across the ground, not as random particle noise. Cloth motion, dust and loose pebbles should reinforce the camera-facing wind direction.
- Before the punch, only subtle pebble vibration/atmosphere is allowed. The decisive cracks, dirt displacement and ground shockwave begin at the punch impact. The huge blue spherical energy wave happens later from the monolith core and must remain visually distinct.
- Camera shake should be physically motivated and low-frequency. Never use constant random shake.
- The punch impact is the physical trigger for the ground fracture. Time the fist contact, ground deformation, dust ring, debris impulse and camera impulse to the same animation event/marker.
- The punch shockwave travels primarily through the ground as cracks, displaced dirt and pebble impulses. It should NOT look like the later luminous spherical TrustChain energy wave.
- Use contact-driven particles and a radial ground impulse. The shockwave should race ahead of the man toward the future monolith emergence point.
- Immediately after the punch, use a short visual beat where the user sees the consequence before the monolith begins its heavy rise.
- Scroll velocity may influence dust intensity and impact strength, but clamp the effect to prevent violent spikes from fast mouse wheels.

6.3 Monolith emergence

- The monolith rises from below the terrain only AFTER the man punches the floor and the physical ground shockwave fractures the area ahead of him. This cause-and-effect sequence is mandatory.
- The rise must feel heavy: slow acceleration, deep rumble visual cues, lagging debris, dust displacement and subtle camera impulses.
- During the rise, the monolith is animation/kinematic controlled for deterministic timing.
- After it settles, it becomes physically reactive or receives equivalent restoring-force behavior so chain tension can affect it slightly.

- It must never become a light object. Chain pulls should create small tilt/shift only, followed by a slow return.

6.4 Instant core ignition + spherical energy wave

- After the monolith settles, briefly reduce scene activity to create anticipation.
- The TrustChain core/logo should then ignite almost instantly with a powerful cyan/blue flash - not a slow fade.
- That ignition emits a giant spherical energy wave centered on the monolith.
- Represent the wave as a translucent expanding sphere/ring field with refraction/distortion, light, particles and local reactions - not as a flat 2D circle.
- The wave is the cause of the environment awakening.

6.5 Rocks energize as the wave reaches them

- Rocks should not all float simultaneously.
- Each rock reacts when the expanding energy sphere physically reaches its position.
- Reaction sequence: surface glow/cracks -> vibration -> lift -> slow rotation/drift.
- Use hero rocks for important silhouettes and instanced/lightweight debris for depth.
- Anchor rocks/portal rocks tied to the approved chain composition should become more strongly energized.

6.6 Chains - only the approved reference chains, all real physics

Non-negotiable chain rule

Do not add extra decorative chains. Recreate only the major chains visible in the approved hero reference, with approximately the same directions, anchor-rock relationships and final composition. Every visible desktop chain uses real physics.

- Each visible link is a Rapier rigid body or an equivalent real physics body.
- Adjacent links are connected with stable joints/constraints.
- Use simple colliders even if the visible chain mesh is detailed.
- Enable sleeping when chains settle.
- Deploy chains sequentially, not all at once. Each lock impact should feel heavy and distinct.
- A guided lead-link/kinematic attractor can be used during deployment so a physical chain reliably travels toward its lock point without becoming chaotic.
- After deployment, all links remain physically active and interactive in both cinematic and normal modes.

6.7 Logo emerges and becomes the navbar brand

- Do not begin with a normal static navbar logo in the cinematic state.
- After the chain/energy sequence stabilizes, a holographic/energy copy of the TrustChain emblem separates from the monolith core.
- The copy follows a curved, elegant trajectory to the top-left corner.
- During travel it scales down and resolves into the navbar brand: emblem + TrustChain wordmark + small VERIFIABLE AI subtitle.
- The monolith core remains glowing after the navbar brand has formed.

6.8 Hero UI reveal after the entire opening event

- Do not show the main hero headline, CTA buttons or four feature cards at the beginning.
- Reveal them only after monolith rise, core ignition, energy wave, rock levitation, chain binding and logo-to-navbar transition are complete.
- Use blur-to-sharp, mask reveals, light sweeps, restrained perspective and upward movement. Do not use a cheap typewriter effect.
- CTA buttons can emerge from below with slight rotateX/perspective before settling.
- In normal/quick mode, the same elements appear during the ~300-500ms resolve into the final hero state.

7. Final hero state is a living interactive world

The end of the opening is not the end of animation. It is the transition from directed cinematic mode into an ambient interactive world. This remains true even when the opening is skipped with ENTER or ESC.

- Wind continues with non-repeating gust variation.
- Dust continues crossing the environment.
- The man has idle/breathing animation and continuous cloth/cloak movement.
- Floating rocks continue slow independent drift and rotation.
- The monolith core has a slow breathing glow and occasional subtle circuitry pulse.
- All approved chains continue to use live physics.
- Particles, stars and distant objects retain subtle parallax and depth movement.
- Do not run every idle effect on obvious synchronized loops.

7.1 Full chain grabbing with cursor

1. Raycast the pointer against the actual visible chain links.
2. On pointer-down, create/move an invisible kinematic grab body at the world-space hit point.
3. Attach the selected link to that grab body using a stable spring/joint strategy.
4. Project pointer movement onto an intuitive drag plane so the chain follows the cursor in 3D.
5. Neighboring links react naturally through physics; do not hand-animate them.
6. Clamp maximum force, drag distance and velocity to avoid exploding the simulation.
7. On release, detach and let the chain oscillate and settle naturally.

cursor pulls chain
-> chain tension increases
-> anchor rock reacts
-> monolith tilts/shifts slightly
-> core/logo pulses
-> dust/debris reacts
-> nearby rocks receive a small response
-> system settles naturally

7.2 Cursor-driven ambient response

- Use subtle camera parallax in the hero only; keep maximum movement restrained.
- Cursor near monolith -> circuitry/core brightness increases slightly.
- Cursor over core/logo -> brief energy pulse through nearby circuit paths.
- Cursor near character -> cloth reacts subtly; optional tiny head/look adjustment, never game-NPC tracking.
- Cursor near floating rocks -> glow and rotation respond slightly.

- Cursor velocity can add a tiny extra impulse to dust/cloak/particles, clamped heavily.
- Interactions should propagate through the world where meaningful instead of behaving like isolated hover effects.

8. Navbar behavior

- Top-left brand appears only after the monolith-logo-to-navbar transition in cinematic mode; in quick mode it appears during the resolve.
- Primary navigation targets: Product, Security, Docs, Pricing.
- Right-side actions: Log In and Get Started.
- Product and Security are internal landing-page navigation targets.
- Docs, Log In and Get Started open a simple Coming Soon page.
- Pricing opens a special cinematic Coming Soon page described later.

8.1 Compact icon-first nav

- The four primary nav targets are initially represented as compact symbols/icons inside a floating glass/nav capsule.
- Hovering or keyboard focusing the capsule expands it horizontally and reveals the labels Product, Security, Docs and Pricing.
- Expansion sequence: border/glow wakes -> capsule stretches -> icons separate -> text resolves -> active item highlight appears.
- On pointer leave, wait briefly before collapsing so the menu does not feel twitchy.
- On mobile/touch, do not rely on hover. Use visible labels or an explicit compact menu control.
- All expansion/collapse must remain accessible with keyboard focus and screen readers.

9. Four glowing hero feature controls

Keep the four approved hero feature concepts as clickable, glowing, animated navigation controls:

Immutable Logs | Merkle Anchored | On-Chain Verified | Publicly Verifiable

- These are not passive badges. Clicking one navigates to its corresponding landing-page section.
- Idle: glassy dark card, cyan edge glow, animated icon, subtle internal particles/scan line.
- Hover: border brightens, icon responds, card lifts slightly and a cursor-following highlight moves across the surface.
- In cinematic mode, click navigation may use a short camera/scene travel transition before arriving at the target section.
- In normal mode, use smooth vertical scroll plus a restrained arrival animation; no giant camera flights.

10. ENTER / ESC Quick View behavior

Finalized meaning of ENTER / ESC

Both keys skip the opening origin cinematic and switch the current page session into NORMAL MODE. They do not open another page, they do not disable 3D, and they do not freeze the hero.

- Resolve the opening to the final hero state in roughly 300-500ms instead of an instantaneous one-frame teleport.

- Place man, monolith, rocks, chains, logo, text, CTAs and feature controls into their final awakened positions.
- Ensure chain bodies/constraints are initialized in a stable final physical state, not visually snapped while physics remains elsewhere.
- Move the internal pinned-scroll progress to the end of the intro so the next user scroll does not replay or reverse the awakening.
- Set introAwakened = true. The intro is one-way for the current session; scrolling back to the top must not bury the monolith again or make the man walk backward.
- Keep wind, dust, cloth, rocks, particles, glow and chain physics alive.
- Switch all later sections to their normal-mode variants: premium animations and contained 3D, but no giant cinematic camera travel.

11. Shared section architecture - content is not duplicated

```
<LandingPage>
  <Hero experienceMode={mode} />
  <ProductSection experienceMode={mode} />
  <ImmutableSection experienceMode={mode} />
  <MerkleSection experienceMode={mode} />
  <BlockchainSection experienceMode={mode} />
  <VerificationSection experienceMode={mode} />
  <RealProductSection experienceMode={mode} />
  <SecuritySection experienceMode={mode} />
  <FinalCTA />
</LandingPage>
```

- The section data/copy source is shared between modes.
- Only animation, layout density, camera behavior and 3D intensity vary by mode.
- Final text for each section will be provided later. Build a clean content configuration/interface now.

12. FULL CINEMATIC MODE - rest of landing page

If the user does not press ENTER or ESC, the rest of the landing page remains a cinematic, scroll-driven journey. It should be inspired by the level of craft found in high-end interactive studio sites such as Lusion and Igloo, without copying their design, assets or exact interactions.

- Opening remains the most intense moment.
- Lower sections are still cinematic/dynamic but use varied pacing so the page does not become constant noise.
- Every transition should feel like the previous visual transforms into the next concept rather than simply cutting to another rectangular section.

12.1 Product world - enter the TrustChain system

- As the user continues from the final hero, camera approaches the monolith/core and travels through it.
- Transition through blue energy into an abstract internal AI-pipeline world.
- Represent the multi-agent flow as monumental floating structures/nodes, not flat cards.
- Scroll moves the camera through the pipeline. Data/energy visibly travels from one stage to the next.
- Text should appear contextually near the current stage using the approved Product copy below.
- Cursor adds subtle parallax and local energy response.

APPROVED PRODUCT COPY - use in both modes

Every AI decision, tracked from first thought to final answer.

TrustChain wraps your AI agents in a four-stage pipeline that records what they did, checks it, scores it, and reports it. Every step is written to Postgres and cryptographically anchored on-chain in the same instant it happens - so the record can't be quietly edited after the fact, even by TrustChain itself.

01 Researcher

Gathers the inputs, context, and evidence behind every agent decision - the raw material an audit starts from.

02 Validator

Checks that output against your rules and facts, flagging anything that doesn't hold up before it moves forward.

03 Scorer

Assigns a quantitative trust score to the run, based on accuracy, consistency, and adherence to process.

04 Reporter

Compiles the full record into a verifiable report - hashed, timestamped, and queued for on-chain anchoring.

Content lock: keep Researcher / Validator / Scorer / Reporter exact; they match the actual LangGraph node names in the running system.

12.2 Product -> Immutable Logs transition

- Outputs/fragments from the pipeline become persistent event objects.
- Those event objects fly through the scene and assemble into a massive futuristic ledger/record structure.
- The transition should make it visually obvious that agent activity becomes an audit trail.

12.3 Immutable Logs world

- Audit events arrive and lock into a large permanent structure with heavy mechanical/energy feedback.
- Movement should feel slow and irreversible.
- Individual foreground records may react on hover, but the overall structure communicates permanence rather than playfulness.

APPROVED IMMUTABLE LOGS COPY

A record that can't be quietly changed.

TrustChain writes every agent step to Postgres and queues it for anchoring in the same database transaction - a crash between "recorded" and "queued" is impossible; either both happen or neither does. Nothing about the audit trail depends on trusting our servers to keep it honest.

- Every step is durably recorded before an agent's run even finishes.
- The step and its anchor entry are written atomically - no window where one exists without the other.
- Tampering after the fact doesn't erase evidence - it creates a detectable mismatch against the anchored proof.

- Use the approved Immutable Logs copy included immediately below; keep the visual layout flexible enough for this copy in both cinematic and normal modes.

12.4 Immutable -> Merkle transition

- The ledger does not simply disappear. It disassembles into luminous hash/data elements.
- Those elements spread into 3D space and reorganize into a Merkle-tree structure.

12.5 Merkle Anchored world

- Build a large 3D Merkle tree from leaf nodes upward as the user scrolls.
- Connections draw/light as parent hashes are formed.
- The user should visually understand many records collapsing into one root without needing a long explanation.
- At the end, the tree collapses visually into a single bright Merkle-root object.
- Use cursor parallax/rotation subtly; do not make it a toy.

APPROVED MERKLE ANCHORED COPY

Thousands of records, one provable root.

Instead of anchoring every step individually, TrustChain batches recent steps into a Merkle tree and anchors only the resulting root on-chain. Each step still gets its own cryptographic proof - a short path of hashes tying it back to that root.

- Steps are batched on a timer, not per-action, keeping anchoring cost predictable.
- Each step's Merkle proof can be checked independently, without re-anchoring anything.
- A single altered step changes its hash, which changes the root - the tampering surfaces anywhere in the tree.

12.6 Merkle -> blockchain transition

- The isolated Merkle root remains in darkness for a short beat, then launches forward as the user scrolls.
- Camera follows it through a portal/energy corridor into the blockchain universe.

12.7 On-chain blockchain universe - major spectacle

Movie-scale blockchain direction

The blockchain world should feel like zero-gravity deep space: huge block objects floating at multiple depths, some close to camera and some extremely distant. Avoid flat "block -> block -> block" diagrams.

- Near-camera blocks are detailed 3D structures with circuitry, internal light and slow rotation.
- Far blocks use instancing/LOD/lightweight geometry to create a vast universe without destroying performance.
- Blocks drift in space, connect via energy/data links and have independent motion.
- The Merkle root enters a selected block and becomes anchored.
- Anchoring triggers a network-wide verification/energy pulse that propagates through connected blocks around the camera.
- Scroll drives forward travel through this world; cursor adds subtle depth parallax.

APPROVED ON-CHAIN VERIFIED COPY

Anchored to a chain, not to us.

The Merkle root is written to a smart contract on Monad. Once it's there, verifying a record doesn't require asking TrustChain to vouch for itself - it requires checking the record's proof against a root that's public, timestamped, and outside our control.

- Anchoring runs as a separate background process, decoupled from the API that serves your data.
- Confirmed roots are reconciled back into the read model by an indexer that watches the chain directly.
- If TrustChain's own database ever disagreed with the chain, the chain is what's authoritative.

12.8 Blockchain -> Public Verification transition

- After the high-energy anchoring event, change emotion: camera pulls far back and the blockchain universe becomes smaller in frame.
- Independent verifier/observer nodes appear around the proof/network.
- Proof paths illuminate one by one, communicating that verification does not depend on trusting TrustChain itself.

12.9 Publicly Verifiable world

- Use a calmer, expansive visual language after the blockchain spectacle.
- Independent nodes connect to the proof, root and on-chain state; successful paths become visually confirmed.

APPROVED PUBLICLY VERIFIABLE COPY

Anyone can check the proof. No account required.

Verification isn't a feature you have to buy access to. Given a step ID and its Merkle proof, any independent party - an auditor, a regulator, a customer - can recompute the hash path and confirm it resolves to the exact root anchored on-chain, using nothing but public chain data.

- Verification logic is small enough to reimplement independently, without trusting TrustChain's own verifier.
- The same proof format is used internally and externally - nothing is held back for outside checks.
- This is what makes the audit trail evidence, not just a claim.
- Use the approved Publicly Verifiable copy included immediately below.

12.10 Cinematic -> real product transition

- A verifier/proof object approaches the camera and morphs into a clean product interface panel.
- This is the credibility pivot from abstract cinematic metaphor to real software.
- The product UI should still animate: rows/statuses/proof paths can resolve sequentially, but the camera should calm down.
- For the real-product visualization, use only minimal realistic UI labels/data needed to demonstrate the interface unless additional product-dashboard copy is supplied later.

12.11 Security world

- Transition into a controlled, dark technical environment rather than another explosion.
- Reintroduce the core/monolith or a simplified central trust object.

- As the user scrolls, multiple protection/isolation layers materialize around it like an exploded engineering diagram.
- Each layer can have a distinct material/energy behavior and subtle cursor separation/reveal.
- Map the four approved Security concepts below to the four protection/isolation layers; do not invent replacement concepts.

APPROVED SECURITY COPY + 4-CARD GRID

Built to survive a compromised database.

TrustChain's audit trail doesn't ask you to trust TrustChain. Every layer below is designed so tampering is detectable, tenants stay isolated, and a proof can be checked independently of our own infrastructure.

Tamper Detection

Every step is hashed and included in a Merkle tree anchored on-chain. An altered record produces a mismatch that is mathematically provable - not just logged.

Atomic Durability

Each audit step and its anchor entry are written in the same database transaction. A crash mid-write can never leave a step silently unrecorded.

Tenant Isolation

Every tenant's data is enforced at two independent layers - application-level filtering and Postgres Row-Level Security - so one missed check can't leak across organizations.

Resilient Infrastructure

Pluggable signing backends (local key, AWS/GCP KMS, HashiCorp Vault) and automatic RPC failover keep anchoring running even when a single provider or key store goes down.

The Security section uses this factual 4-card grid. In cinematic mode, map the four concepts to four physical protection/isolation layers. In normal mode, use the same content in a contained animated layered visualization.

13. NORMAL MODE - same sections, restrained cinematics

If ENTER or ESC was pressed during the opening, the visitor has explicitly chosen a faster browsing experience. Respect that choice. The landing page should still feel premium, animated and unmistakably TrustChain, but remove giant camera flights, long pinned sequences and full-screen world traversal.

Normal-mode rule

Standard vertical scrolling + premium entrance animations + contained 3D visualizations + hover/cursor reactions. No enormous camera shifts, no fly-throughs, no long cinematic detours.

13.1 Product - normal mode

- Use a contained 3D pipeline visualization beside/behind the content.
- Nodes rise/resolve as the section enters the viewport; connections draw and data flows between them.
- Camera remains essentially fixed; cursor only creates light parallax/depth.

13.2 Immutable Logs - normal mode

- Use a contained ledger/record structure that assembles on viewport entry.
- Records can lock sequentially with a small visual seal/flash.

- Hover can reveal subtle depth or record emphasis.

13.3 Merkle - normal mode

- Show a contained interactive 3D Merkle tree.
- On enter: leaves appear -> edges draw -> parents illuminate -> root activates.
- Cursor can slightly rotate/parallax the tree and highlight paths.

13.4 Blockchain - normal mode

- Keep the movie-style floating blocks, but contain them within a section scene.
- Blocks drift/rotate; links appear; a root enters a block; a localized verification pulse propagates.
- No user fly-through and no massive camera relocation.

13.5 Public Verification - normal mode

- Show a central proof/network object with independent verifier nodes around it.
- As the section enters, proof paths resolve sequentially.
- Hovering a verifier highlights its complete verification path.

13.6 Real product - normal mode

- Use normal premium interface animation: depth entrance, status lights, path drawing, subtle card tilt.
- No large camera choreography.

13.7 Security - normal mode

- Use a contained layered security visualization around a central trust object.
- Layers appear on viewport entry and separate subtly on hover.
- Keep it technical and calm.

14. Navigation and destination behavior

Control	Behavior
Product	Smooth-scroll to Product section. Cinematic travel in cinematic mode; normal smooth scroll in normal mode.
Security	Smooth-scroll to Security section with mode-appropriate arrival behavior.
Docs	Open simple Coming Soon page.
View Docs	Open simple Coming Soon page.
Log In	Open simple Coming Soon page.
Get Started	Open simple Coming Soon page.
Pricing	Open dedicated short cinematic Coming Soon page.
Immutable Logs card	Navigate to Immutable Logs section.
Merkle Anchored card	Navigate to Merkle section.

On-Chain Verified card	Navigate to blockchain/on-chain section.
Publicly Verifiable card	Navigate to public verification section.

14.1 Simple Coming Soon page

- Used for Docs, View Docs, Log In and Get Started.
- Plain white or black background, simple TrustChain branding, Coming Soon message and Back/Return action.
- No Three.js world, no large animation, no unnecessary engineering because these pages will be replaced later.

14.2 Pricing Coming Soon - only special placeholder page

- Pricing alone gets a small cinematic placeholder.
- Keep it short: roughly 2-4 seconds, visually tied to TrustChain, then settle on Pricing / Coming Soon.
- Do not build another huge hero; this is a deliberate polished placeholder, not a second experience.
- Provide Return to TrustChain.

15. Smoothness and production-quality motion

Primary engineering objective

The experience must remain smooth under real browser conditions. Cinematic ambition is not an excuse for dropped frames, unstable physics, scroll desynchronization or UI lag.

- Target 60 fps on a typical modern desktop. Degrade gracefully rather than forcing full quality everywhere.
- Smoothness is a release requirement, not optional polish. Do not ship a visually ambitious scene that stutters, desynchronizes scroll and physics, causes layout jumps, or makes camera motion uncomfortable. Reduce visual complexity dynamically before sacrificing motion stability.
- Use React Three Fiber / Three.js for 3D, Rapier for physics, GSAP ScrollTrigger for scroll timelines, and postprocessing selectively.
- A smooth-scroll layer such as Lenis may be used if it integrates cleanly with ScrollTrigger; avoid adding it if native scroll is already smooth and stable.
- Use damped/interpolated camera/object transforms. Avoid direct noisy mapping of wheel delta to camera transforms.
- Clamp scroll velocity influence and physics impulses.
- Never update high-frequency animation state through React setState on every frame. Use refs, frame loops and physics engine state.
- Use instancing, LOD and lightweight materials for the blockchain universe and distant debris.
- Use simple physics colliders. The visible mesh can be detailed; the collider should not be.
- Cap devicePixelRatio dynamically, especially on high-DPI laptops.
- Lazy-load heavy GLBs/textures and display a polished minimal loader if needed.
- Pause/reduce nonessential simulation when the tab is hidden.
- Profile draw calls, shader cost, physics bodies, texture memory and main-thread scripting before final polish.
- Use physically motivated camera impulses rather than a continuously shaking camera.

- Postprocessing should be restrained: controlled bloom, fog/atmosphere, optional SSAO, very subtle vignette/noise; no overblown gaming filter.

15.1 Scroll timeline stability

- Opening progress must be deterministic from scroll progress even if the user scrolls rapidly, reverses slightly or uses a trackpad.
- However, once introAwakened becomes true, do not reverse the world back into its pre-awakening state.
- Use timeline labels / normalized progress ranges rather than scattering magic scroll numbers through components.
- Keep physical chain simulation decoupled from scroll after deployment so user interaction remains natural.
- When Quick View fast-forwards the intro, synchronize scroll state, GSAP timeline state, physics initialization and DOM visibility in the same transition.

15.2 Mode switching stability

- Switching cinematic -> normal must not remount the entire page if avoidable.
- Prefer variant props/state and gracefully cancel/complete pinned timelines.
- Do not leave invisible pinned spacers or locked body scrolling after Quick View.
- Ensure focus remains sensible after mode switch and keyboard controls are restored.

16. Recommended implementation stack

Existing React / Next.js / Vite stack (preserve repo choice)
 three
 @react-three/fiber
 @react-three/drei
 @react-three/rapier
 gsap + ScrollTrigger
 @react-three/postprocessing
 optional: lenis (only if it improves scroll quality without conflicts)

- Use TypeScript if the repository is TypeScript.
- Keep route/UI content separate from 3D scene logic.
- Use a central asset manifest so procedural placeholders can later be replaced by final Blender GLB assets without rewriting the scene architecture.

17. Asset strategy

- Preferred final assets: environment/terrain GLB, rigged character GLB, monolith GLB, chain-link GLB, hero rocks/anchor rocks GLB, optimized textures.
- If a final asset is missing, do not block implementation. Build procedural placeholders that preserve the final architecture and can be swapped later.
- Do not use the approved hero PNG as the final interactive background. It is a composition/art-direction reference.
- Character: rigged walk, stop, brace/react, look-up and idle clips are preferred.
- Cloth: prefer bone/secondary animation or lightweight shader/vertex motion rather than expensive real-time full cloth simulation.

18. Mobile, accessibility and reduced motion

- Desktop gets the complete cinematic + physics experience.
- Tablet reduces particle density/postprocessing and may simplify physics update cost.
- Mobile prioritizes readability. Keep core visuals but simplify chain detail, particles, shadows and camera behavior as required.
- Do not rely on hover-only navigation on touch devices.
- Honor prefers-reduced-motion by starting close to the normal-mode/final-hero experience, reducing shake and disabling major camera travel.
- Keep semantic HTML, keyboard focus, link/button semantics, text selection and screen-reader usability intact.
- Do not let WebGL block scrolling or trap focus.

19. Suggested code organization

```
src/landing/  
  TrustChainLandingPage.*  
  config/  
    content.*           // final copy injected later  
    assets.*  
    motion.*  
    performance.*  
  components/  
    Navbar.*  
    HeroUI.*  
    FeatureNavCards.*  
    ProductSection.*  
    ImmutableSection.*  
    MerkleSection.*  
    BlockchainSection.*  
    VerificationSection.*  
    RealProductSection.*  
    SecuritySection.*  
    ComingSoonPage.*  
    PricingComingSoonPage.*  
  three/  
    HeroScene.*  
    CinematicCamera.*  
    Character.*  
    Monolith.*  
    PhysicalChain.*  
    ChainGrabController.*  
    FloatingRocks.*  
    SphericalEnergyWave.*  
    DustAndWind.*  
    HeroPostProcessing.*  
    ProductWorld.*  
    LedgerWorld.*  
    MerkleWorld.*  
    BlockchainWorld.*  
    VerificationWorld.*  
    SecurityWorld.*  
  hooks/  
    useExperienceMode.*  
    useOpeningTimeline.*
```

20. Implementation phases - build and verify incrementally

8. Phase 0 - Inspect the existing repo, then remove/replace only the old landing-page visual implementation. Preserve underlying infrastructure.
9. Phase 1 - Build the new DOM shell, routes, content config placeholders, full-screen canvas, responsive nav skeleton and final hero layout.
10. Phase 2 - Build hero terrain, camera, lighting, atmosphere, character placeholder/asset and monolith.
11. Phase 3 - Implement the behind-the-man camera follow, punch wind-up/impact, ground shockwave, fractured terrain and deterministic monolith emergence triggered by the punch.
12. Phase 4 - Implement instant core ignition, spherical energy wave and wave-triggered rock levitation.
13. Phase 5 - Recreate only approved chains with full Rapier link physics, sequential deployment and stable attachment.
14. Phase 6 - Implement logo emergence -> navbar transformation, hero copy/CTA/card reveal, icon-first expanding navbar.
15. Phase 7 - Implement live hero interactions: chain grab, monolith reaction, cursor-responsive glow, wind, dust, cloth and rock behavior.
16. Phase 8 - Implement ENTER/ESC Quick View: 300-500ms resolve, one-way awakening, mode switch and stable scroll/physics synchronization.
17. Phase 9 - Implement cinematic-mode Product -> Immutable -> Merkle -> blockchain -> verification -> real product -> Security world transitions.
18. Phase 10 - Implement normal-mode variants of the same sections with contained 3D and premium entrance/hover animation.
19. Phase 11 - Implement internal section navigation and four glowing feature controls.
20. Phase 12 - Implement simple Coming Soon routes plus special Pricing cinematic Coming Soon.
21. Phase 13 - Mobile/reduced-motion variants, performance profiling, asset optimization, accessibility and final polish.

21. Acceptance criteria

- **[PASS]** Existing landing-page visual UI is replaced; backend/app infrastructure remains intact.
- **[PASS]** Opening is driven by scroll, not a forced autoplay movie.
- **[PASS] The man appears first, camera follows from behind, he visibly punches the ground, and that impact triggers the ground shockwave before the monolith rises.**
- **[PASS]** The monolith visibly emerges from below the ground.
- **[PASS]** The core/logo ignites abruptly and emits a large spherical energy wave.
- **[PASS]** Rocks energize and levitate when the expanding wave reaches them, not all at once.
- **[PASS]** Only chains matching the approved reference composition are present.
- **[PASS]** Every visible desktop chain uses real physics and remains interactive after the opening.
- **[PASS]** Cursor can grab a chain and the physical response propagates through links, anchor and monolith.
- **[PASS]** Logo separates from the monolith and resolves into the top-left navbar brand.
- **[PASS]** Hero text, CTAs, navbar controls and four feature cards appear only after the cinematic awakening is complete (or during Quick View resolve).
- **[PASS]** Final hero remains alive: wind, dust, cloth, rocks, glow, particles and chain physics continue.
- **[PASS]** Navbar four primary items are compact/icon-first and expand to show labels on hover/focus.

- **[PASS]** Product and Security navigate to landing-page sections.
- **[PASS]** Four glowing feature cards navigate to their matching sections.
- **[PASS]** Docs, View Docs, Log In and Get Started open a plain Coming Soon page.
- **[PASS]** Pricing opens the dedicated short cinematic Coming Soon page.
- **[PASS]** ENTER and ESC during the opening both resolve to the final hero and activate normal mode.
- **[PASS]** Quick View does not freeze Three.js or hero interactions.
- **[PASS]** Quick View does not cause the intro to reverse if the user scrolls upward later.
- **[PASS]** Cinematic mode keeps the full connected Product/Logs/Merkle/blockchain/verification/Security journey.
- **[PASS]** Normal mode uses the same sections with smaller contained animations and no giant camera travel.
- **[PASS]** The blockchain cinematic world uses floating movie-scale blocks in space with depth, drift and network energy.
- **[PASS]** Section copy is supplied through a centralized config and is not hardcoded across animation components.
- **[PASS]** Mobile/touch and prefers-reduced-motion behaviors are usable and deliberate.
- **[PASS]** Desktop performance is profiled and obvious frame-rate, physics or memory bottlenecks are fixed.
- **[PASS]** No scroll lock, pinning bug, invisible spacer, focus trap or broken route remains after mode switches or navigation.

22. Final instruction to Codex / coding agent

IMPLEMENT THIS SPECIFICATION. Do not merely describe it.

Before coding:

1. Inspect the repository and identify the current frontend framework, routes, styling system and reusable infrastructure.
2. Identify exactly which existing landing-page UI files can be removed/replaced without touching unrelated product/backend code.
3. Produce a short phased plan mapped to this specification.
4. Then execute phase by phase, running the app and fixing errors after every major phase.

Important: Do not invent final marketing body copy for the landing-page sections. Use a centralized content config with temporary placeholders. The final section content will be supplied in a later revision. Focus this implementation on architecture, physics, interaction, cinematics, navigation, responsiveness and smoothness.

23. Content status for this FINAL implementation prompt

Approved copy for Product, Security, Immutable Logs, Merkle Anchored, On-Chain Verified and Publicly Verifiable is now integrated into this specification and should be implemented from the centralized content configuration. Do not redesign the motion architecture around copy; keep content data-driven.

Remaining flexible content areas: hero supporting copy may use the current approved hero composition wording; real product/verified-run UI may use concise realistic labels/data; final CTA/footer microcopy may be chosen to fit the existing TrustChain product language. Do not rewrite or rename the approved section copy above unless explicitly instructed later.